MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, TRANSLATOR, DIGITAL, KEY CODE DLB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital translator driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.060 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DLB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/312-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc and (V_{DD}) of -5.0 ±0.25 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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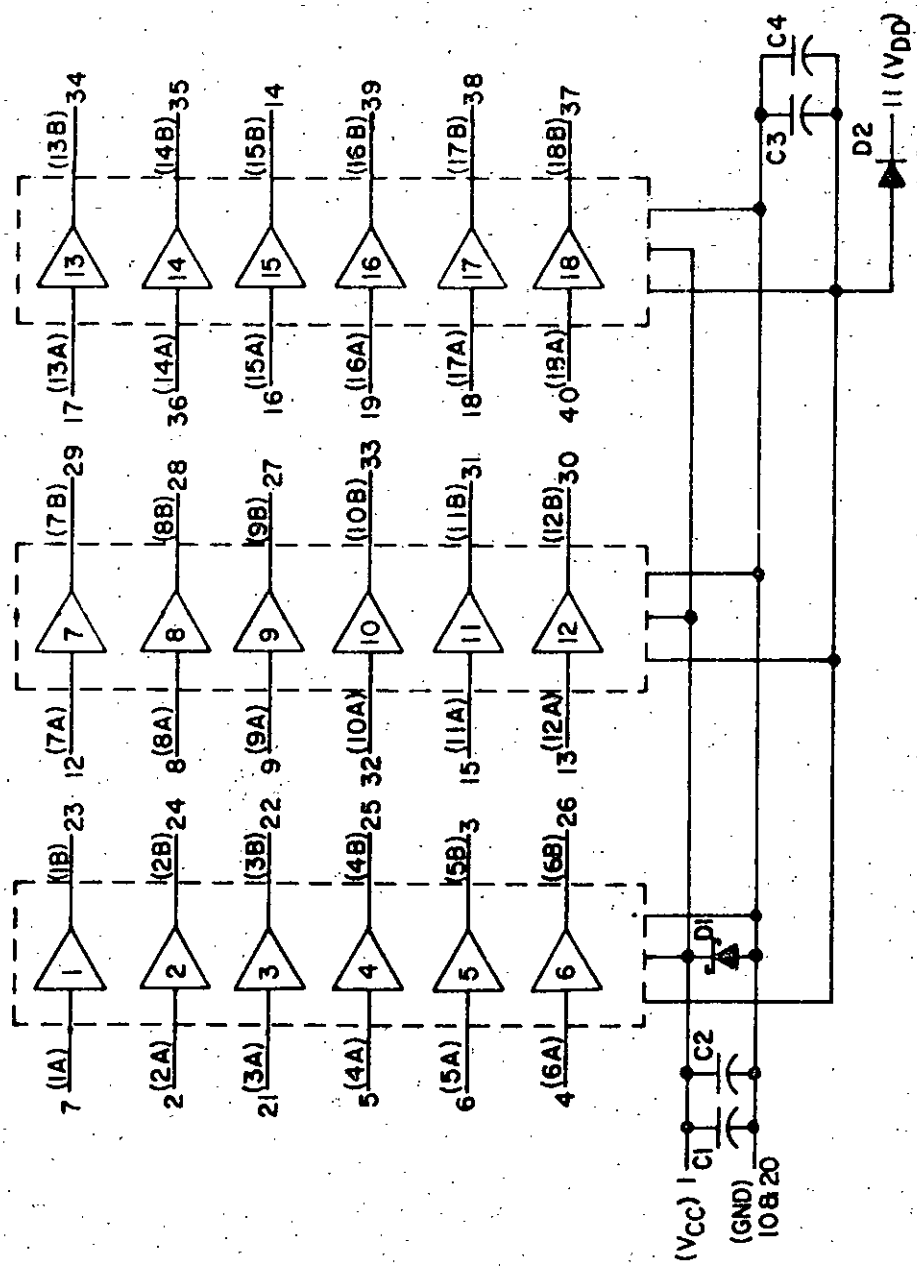


FIGURE 1. DLB module function diagram, translator driver.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	Input breakdown current	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
B	Isolation	IV	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
C1, C2	Power sequencing	V	S	Module must pass all remaining tests								
D1-D18	Input capacitance	VI	Q	2	15	2	15	2	15	2	15	pF
E1-E18	Output short circuit current	VII	X	25	310	25	310	25	310	25	310	mA
F1-F18	Static output voltage, high level	VIII	X	0.0	-0.5	0.0	-0.5	0.0	-0.5	0.0	-0.5	V
G1-G18	Static output voltage, low level	IX	X	-3.0	-5.0	-3.0	-5.0	-3.0	-5.0	-3.0	-5.0	V
H1-H18	Input current, low level	X	X	-0.05	-1.73	-0.05	-1.73	-0.05	-1.73	-0.05	-1.73	mA
J1-J18	Input current, high level	XI	X	0	40	0	40	0	40	0	40	μA
K1	Power supply current, high level, V_{CC}	XII	S	30	120	30	120	30	120	30	120	mA
K2	Power supply current, high level, V_{DD}	XII	S	30	185	30	185	30	185	30	185	mA
K3	Power supply current, low level, V_{CC}	XII	S	30	120	30	120	30	120	30	120	mA
K4	Power supply current, low level, V_{DD}	XII	S	30	140	30	140	30	140	30	140	mA
L1-L18	Transition time (t_{TLH})	XIII	S	10	165	10	165	10	165	10	165	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M18	Transition time (t_{THL})	XIV	S	10	185	10	185	10	185	10	185	ns
N1-N18	Propagation delay time (t_{PLH})	XV	S	10	200	10	200	10	200	10	200	ns
P1-P18	Propagation delay time (t_{PHL})	XVI	S	10	200	10	200	10	200	10	200	ns
Q1	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
Q2	Power supply transient amplitude	XVII	Q	0	500	0	500	0	500	0	500	mV
R1-R18	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
S1-S18	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient rise time	XX	S	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0	ms
U1-U18	Output impedance check	XXI	Q	0	70	0	70	0	70	0	70	µA

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

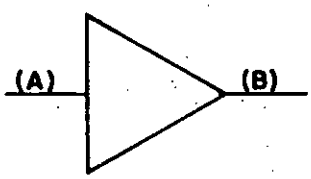
6.3 Module function. The translator driver module consists of 18 interface line driver circuits. It is designed to drive up to a maximum of 300 feet of 2U cable (26 AWG twisted shielded pair) while also translating from positive to negative voltage levels. When used with twisted shielded pair, the signal side of the line should be connected to the module output, the low side of the line to signal ground, and the shield to chassis. The DLB module is intended for use with translator receiver module DLC (MIL-M-28787/313), or its equivalent. The combination is noninverting (from positive voltage levels DLB inputs to positive voltage levels of DLC output). Each driver circuit will withstand, without failure, a short between the driver output and ground. With power to the module removed, the driver output will present a minimum impedance of 100 kilohms to a negative voltage on the line between -3 and -7 V.

6.3.1 Input to output relationship. Each driver shall have the following input and output relationship for positive logic:

TABLE II. Input and output for positive logic.

Logic function: A = B

A	B
L	L
H	H



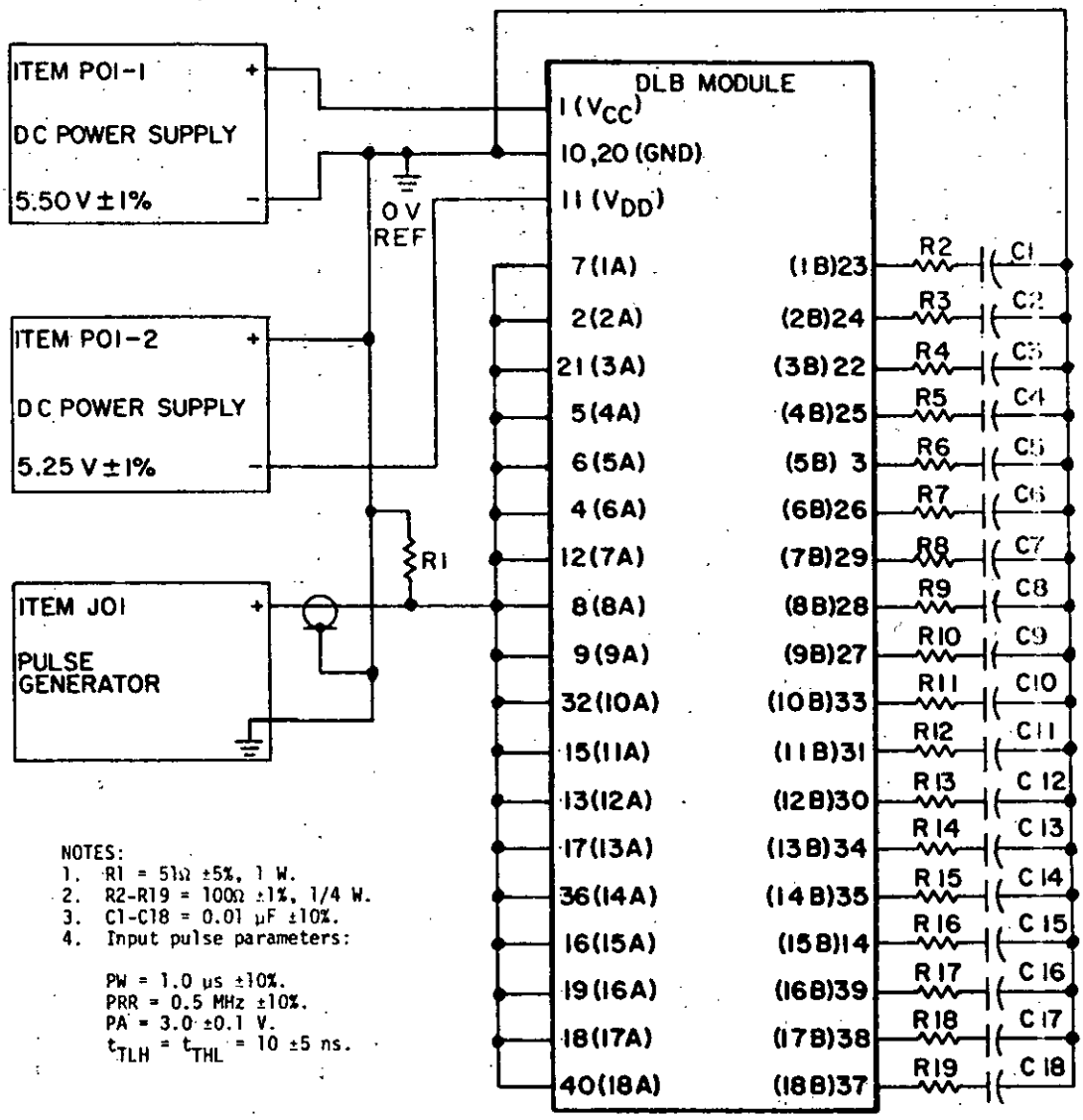
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Preparing activity:
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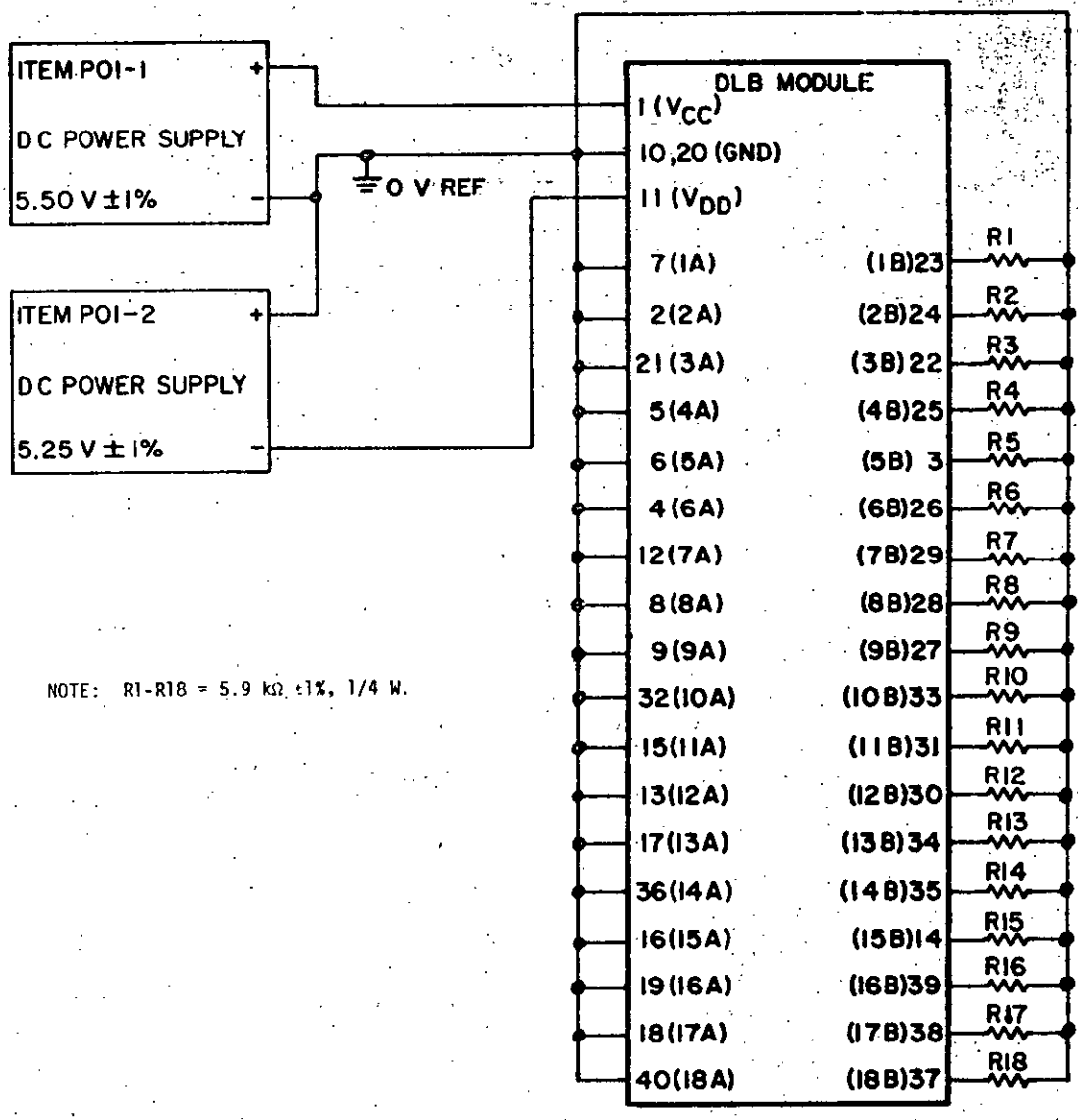


NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2-R19 = 100Ω ±1%, 1/4 W.
3. C1-C18 = 0.01 µF ±10%.
4. Input pulse parameters:

   PW = 1.0 µs ±10%.
   PRR = 0.5 MHz ±10%.
   PA = 3.0 ±0.1 V.
   $t_{TLH} = t_{THL}$ = 10 ±5 ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R18 = 5.9 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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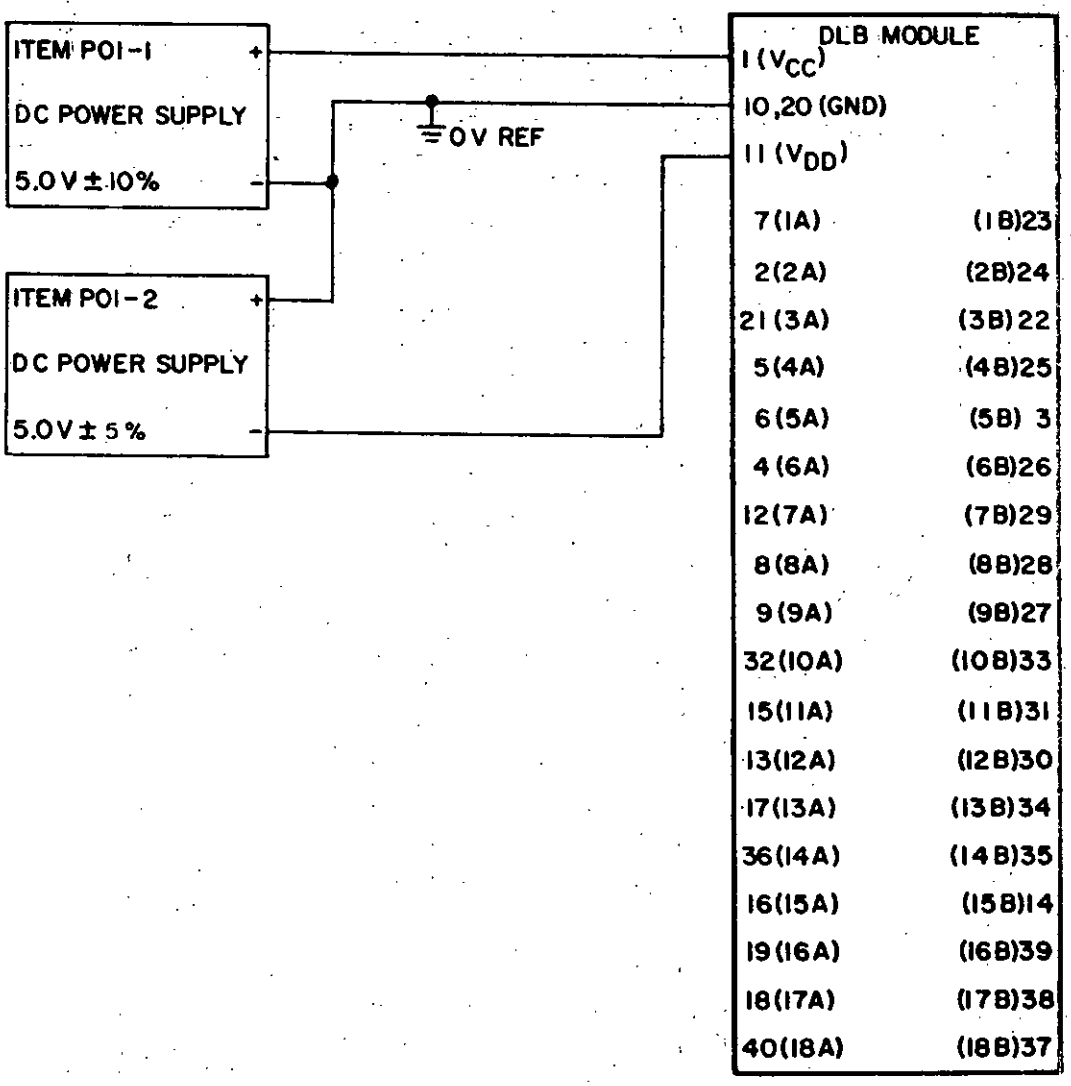


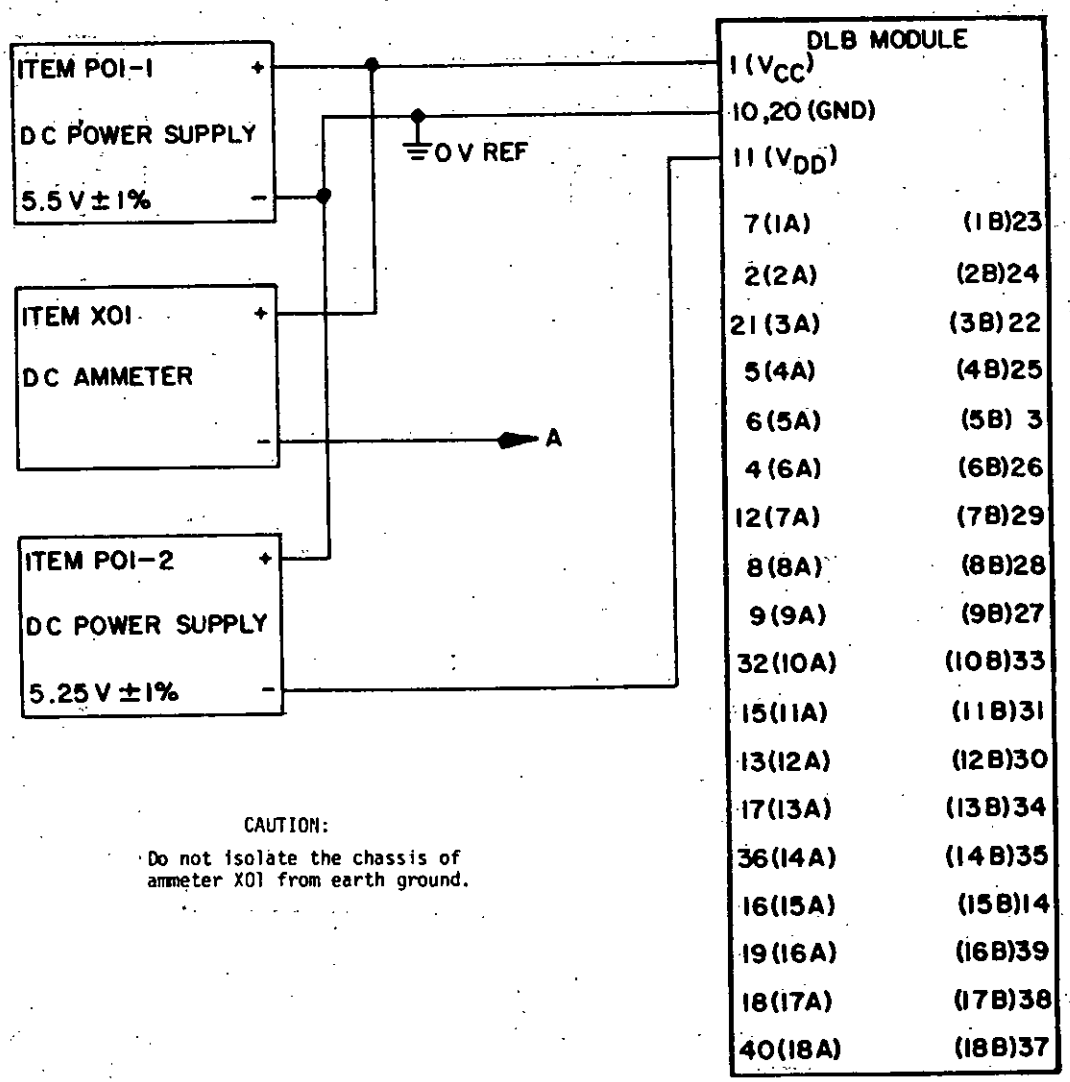
FIGURE 4. Vibration test circuit.

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TABLE III. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
A1	1A Input breakdown current	7	0.0	1.0	mA	5
A2	2A	2				
A3	3A	21				
A4	4A	5				
A5	5A	6				
A6	6A	4				
A7	7A	12				
A8	8A	8				
A9	9A	9				
A10	10A	32				
A11	11A	15				
A12	12A	13				
A13	13A	17				
A14	14A	36				
A15	15A	16				
A16	16A	19				
A17	17A	18				
A18	18A	40				

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CAUTION:
Do not isolate the chassis of ammeter XOI from earth ground.

FIGURE 5. Input breakdown current test circuit for table III.

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TABLE IV. Isolation test.

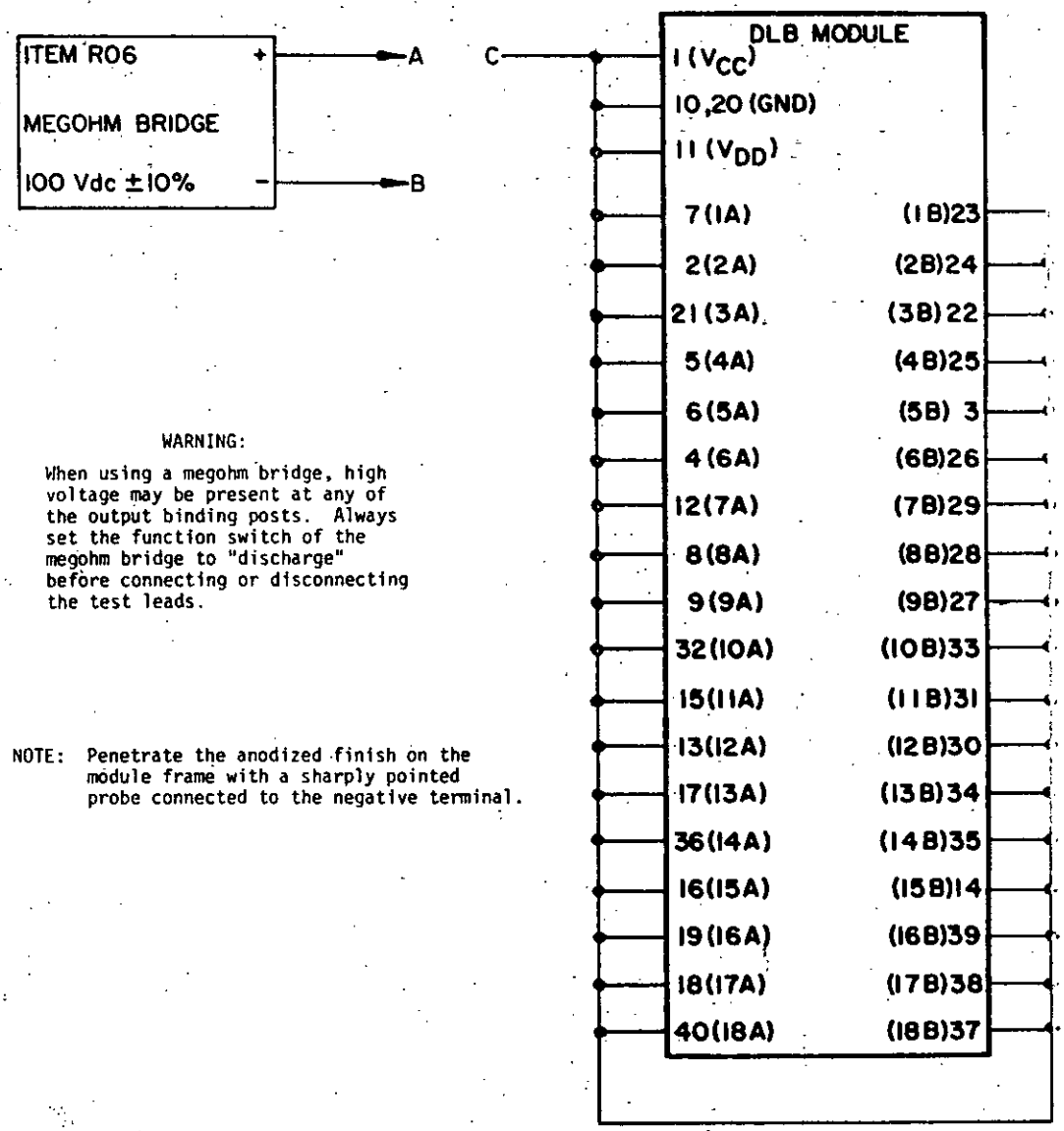
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Isolation	C	Frame	10	∞	MΩ	6

TABLE V. Power sequencing test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Power sequencing	1	Open	1/	1/	2/	7
C2	Power sequencing	Open	11	1/	1/	2/	7

1/ The module shall pass all remaining tests.
2/ The test connections shall be made for a maximum of 60 seconds.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table IV.

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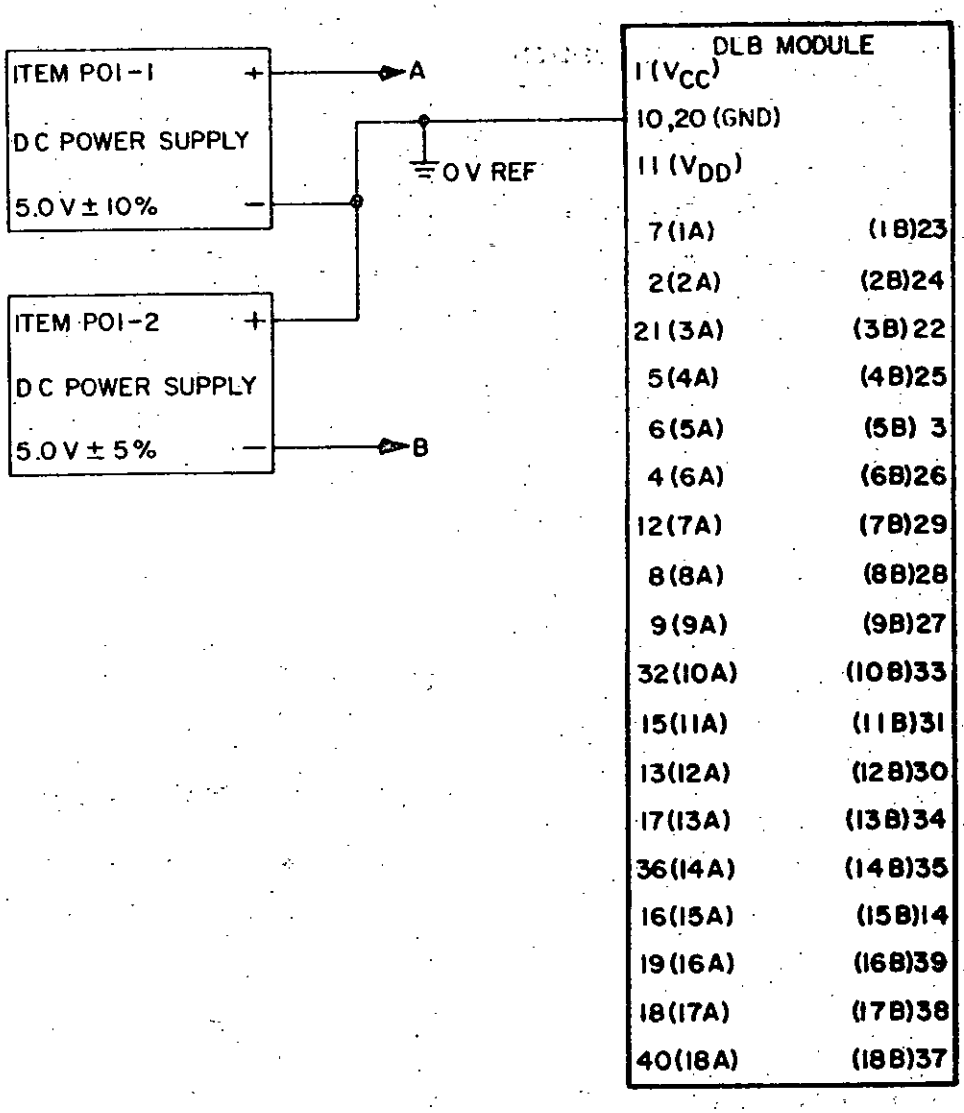


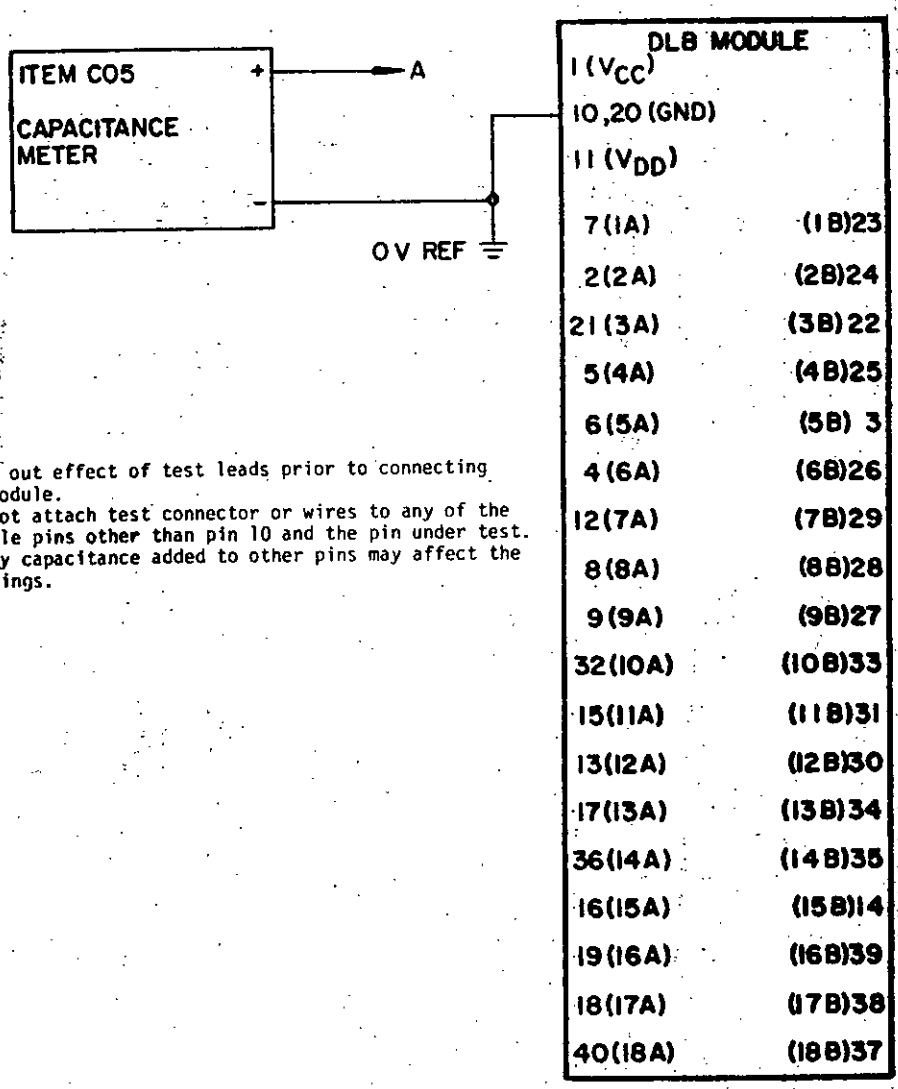
FIGURE 7. Power sequencing test circuit for table V.

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TABLE VI. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1A Input capacitance	7	2	15	pF	8
D2	2A	2				
D3	3A	21				
D4	4A	5				
D5	5A	6				
D6	6A	4				
D7	7A	12				
D8	8A	8				
D9	9A	9				
D10	10A	32				
D11	11A	15				
D12	12A	13				
D13	13A	17				
D14	14A	36				
D15	15A	16				
D16	16A	19				
D17	17A	18				
D18	18A	40				

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NOTES:
1. Null out effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

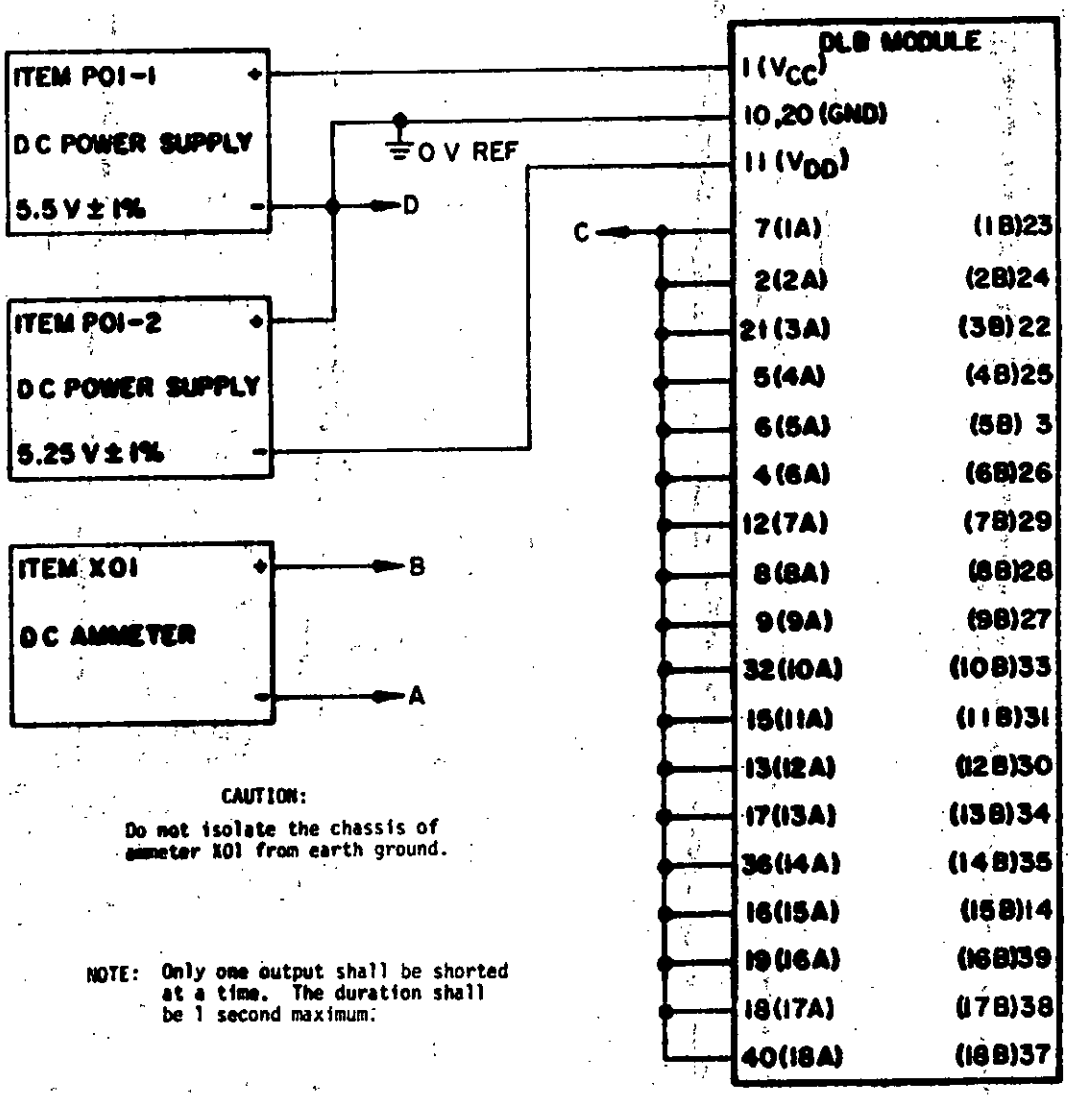
FIGURE 8. Input capacitance test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1B Output short circuit current	23	D	D	25	310	mA	9
E2	2B	24						
E3	3B	22						
E4	4B	25						
E5	5B	3						
E6	6B	26						
E7	7B	29						
E8	8B	28						
E9	9B	27						
E10	10B	33						
E11	11B	31						
E12	12B	30						
E13	13B	34						
E14	14B	35						
E15	15B	14						
E16	16B	39						
E17	17B	38						
E18	18B	37						

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Only one output shall be shorted at a time. The duration shall be 1 second maximum.

FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Static output voltage (high-level) test.

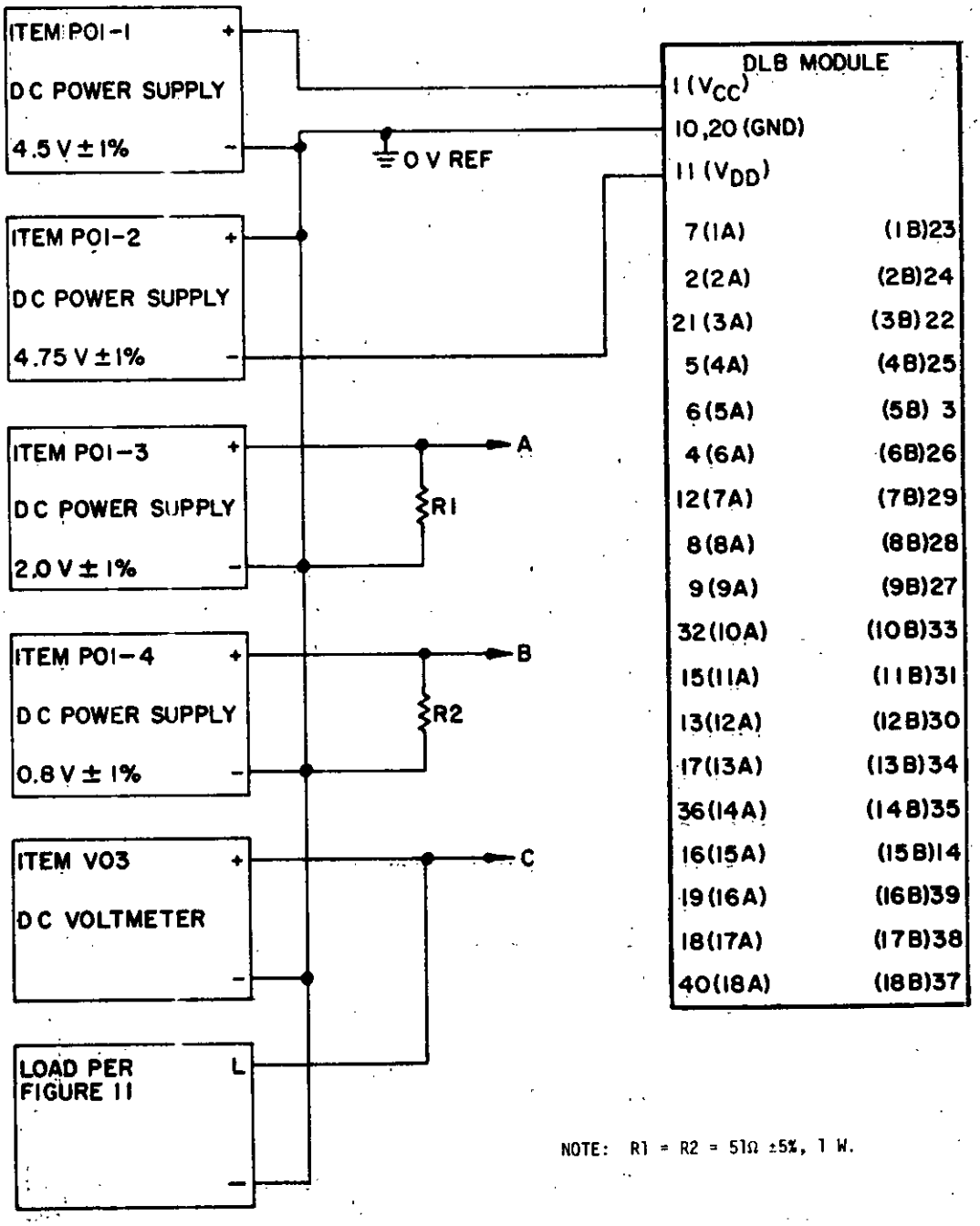
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1B Static output voltage, high level	7	Open	23	0.0	-0.5	V	10
F2	2B	2		24				
F3	3B	21		22				
F4	4B	-5		25				
F5	5B	6		3				
F6	6B	4		26				
F7	7B	12		29				
F8	8B	8		28				
F9	9B	9		27				
F10	10B	32		33				
F11	11B	15		31				
F12	12B	13		30				
F13	13B	17		34				
F14	14B	36		35				
F15	15B	16		14				
F16	16B	19		39				
F17	17B	18		38				
F18	18B	40		37				

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TABLE IX. Static output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1B Static output voltage, low level	Open	7	23	-3.0	-5.0	V	10
G2	2B		2	24				
G3	3B		21	22				
G4	4B		5	25				
G5	5B		6	3				
G6	6B		4	26				
G7	7B		12	29				
G8	8B		8	28				
G9	9B		9	27				
G10	10B		32	33				
G11	11B		15	31				
G12	12B		13	30				
G13	13B		17	34				
G14	14B		36	35				
G15	15B		16	14				
G16	16B		19	39				
G17	17B		18	38				
G18	18B		40	37				

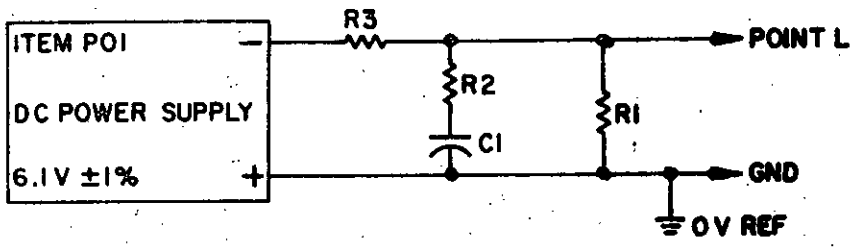
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 10. Static output voltage (high-level and low-level) test circuit for tables VIII and IX.

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NOTES:
1. C1 = .01 µF ±10%.
2. RI = 2.05 kΩ ±1%, 1/4 W.
3. R3 = 3.16 kΩ ±1%, 1/4 W.
4. R2 = 100Ω ±1%, 1/4 W.
5. R2 and C1 are not required for dc electrical tests F1-F18 and G1-G18.
6. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, item POI shall have a maximum overshoot of 500 mV.

FIGURE 11. DLB logic load circuit.

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TABLE X. Input current (low-level and high-level) test.

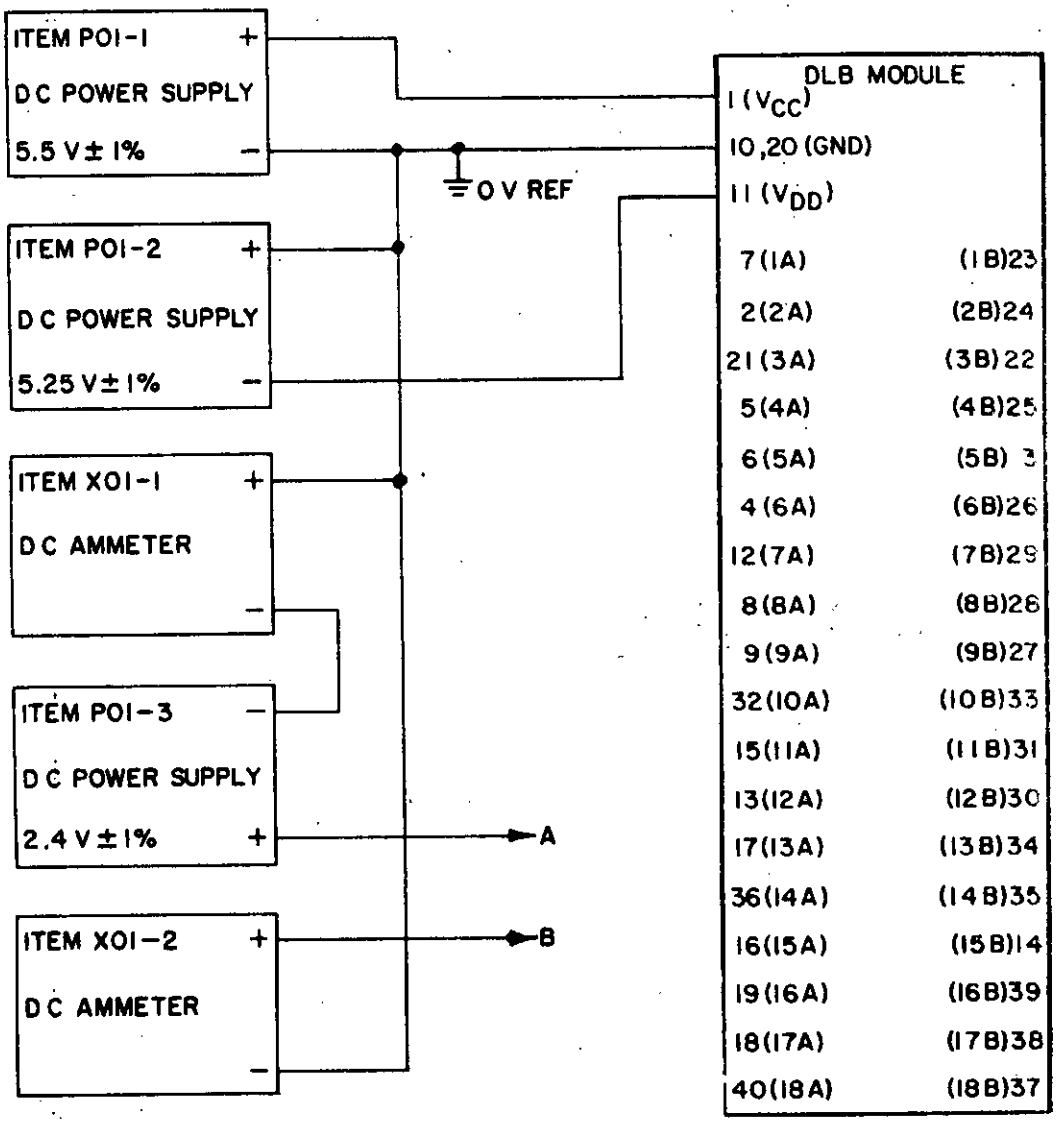
Test condition		Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
:::	1A	Input current, low level	Open	7	-0.05	-1.73	mA	12
H2	2A			2				
H3	3A			21				
H4	4A			5				
H5	5A			6				
H6	6A			4				
H7	7A			12				
H8	8A			8				
H9	9A			9				
H10	10A			32				
H11	11A			15				
H12	12A			13				
H13	13A			17				
H14	14A			36				
H15	15A			16				
H16	16A			19				
H17	17A			18				
H18	18A			40				

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TABLE XI. Input current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	7	Open	0	40	µA	12
J2	2A	2					
J3	3A	21					
J4	4A	5					
J5	5A	6					
J6	6A	4					
J7	7A	12					
J8	8A	8					
J9	9A	9					
J10	10A	32					
J11	11A	15					
J12	12A	13					
J13	13A	17					
J14	14A	36					
J15	15A	16					
J16	16A	19					
J17	17A	18					
J18	18A	40					

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since ammeters XOI-1 and XOI-2 are not used simultaneously, only one ammeter XOI is required.
2. With point "A" disconnected, ammeter XOI-1 should indicate zero current on the 1 µA range.

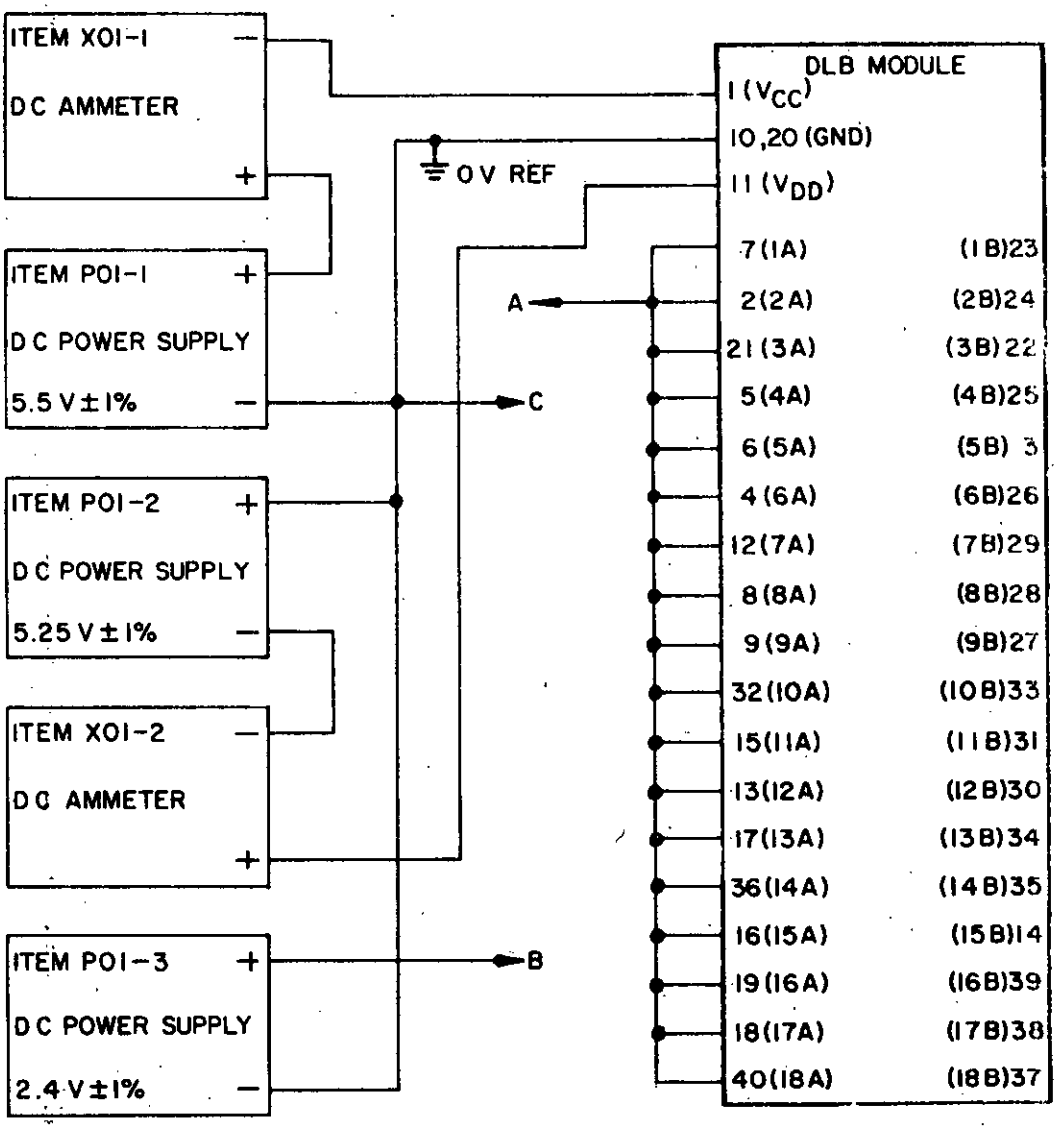
FIGURE 12. Input current (low-level and high-level) test circuit for tables X and XI.

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TABLE XII. V_{CC} and V_{DD} power supply current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
K1	V_{CC} Power supply current, high level	A	Open	30	120	mA	13
K2	V_{DD} Power supply current, high level	A	Open	30	185		
K3	V_{CC} Power supply current, low level	Open	A	30	120		
K4	V_{DD} Power supply current, low level	Open	A	30	140		

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 13. V_{CC} and V_{DD} power supply current (high-level and low-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition		Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	1B	Transition time (t_{TLH})	7	23	10	165	ns	14, 15
L2	2B		2	24				
L3	3B		21	22				
L4	4B		5	25				
L5	5B		6	3				
L6	6B		4	26				
L7	7B		12	29				
L8	8B		8	28				
L9	9B		9	27				
L10	10B		32	33				
L11	11B		15	31				
L12	12B		13	30				
L13	13B		17	34				
L14	14B		36	35				
L15	15B		16	14				
L16	16B		19	39				
L17	17B		18	38				
L18	18B		40	37				

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TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1B Transition time (t_{THL})	7	23	10	185	ns	14, 15
M2	2B	2	24				
M3	3B	21	22				
M4	4B	5	25				
M5	5B	6	3				
M6	6B	4	26				
M7	7B	12	29				
M8	8B	8	28				
M9	9B	9	27				
M10	10B	32	33				
M11	11B	15	31				
M12	12B	13	30				
M13	13B	17	34				
M14	14B	36	35				
M15	15B	16	14				
M16	16B	19	39				
M17	17B	18	38				
M18	18B	40	37				

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TABLE XV. Propagation delay time (t_{PLH}) test.

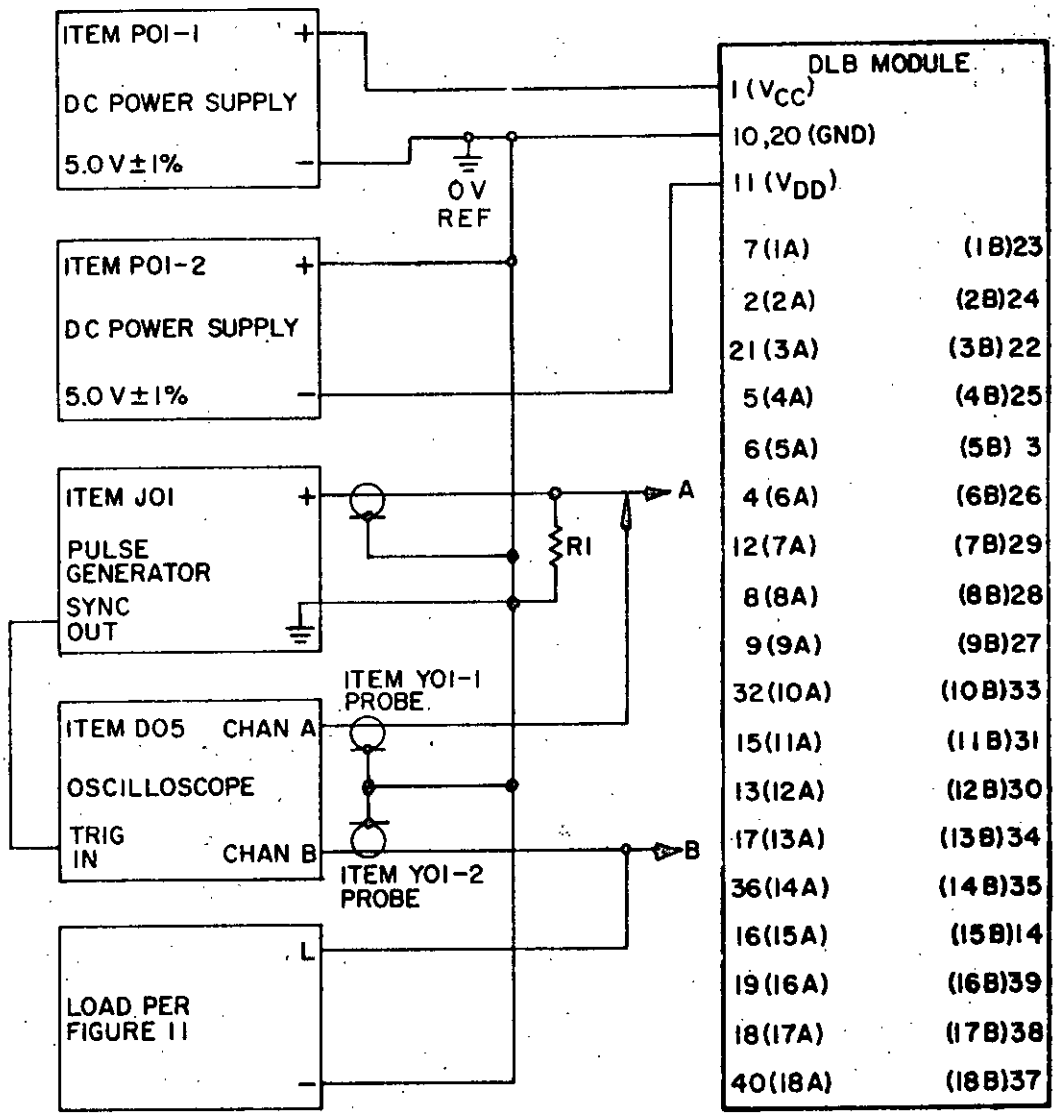
Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	1B	Propagation delay time (t_{PLH})	7	23	10	200	ns	14, 15
N2	2B		2	24				
N3	3B		21	22				
N4	4B		5	25				
N5	5B		6	3				
N6	6B		4	26				
N7	7B		12	29				
N8	8B		8	28				
N9	9B		9	27				
N10	10B		32	33				
N11	11B		15	31				
N12	12B		13	30				
N13	13B		17	34				
N14	14B		36	35				
N15	15B		16	14				
N16	16B		19	39				
N17	17B		18	38				
N18	18B		40	37				

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TABLE XVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	1B	Propagation delay time (t_{PHL})	7	23	10	200	ns	14, 15
P2	2B		2	24				
P3	3B		21	22				
P4	4B		5	25				
P5	5B		6	3				
P6	6B		4	26				
P7	7B		12	29				
P8	8B		8	28				
P9	9B		9	27				
P10	10B		32	33				
P11	11B		15	31				
P12	12B		13	30				
P13	13B		17	34				
P14	14B		36	35				
P15	15B		16	14				
P16	16B		19	39				
P17	17B		18	38				
P18	18B		40	37				

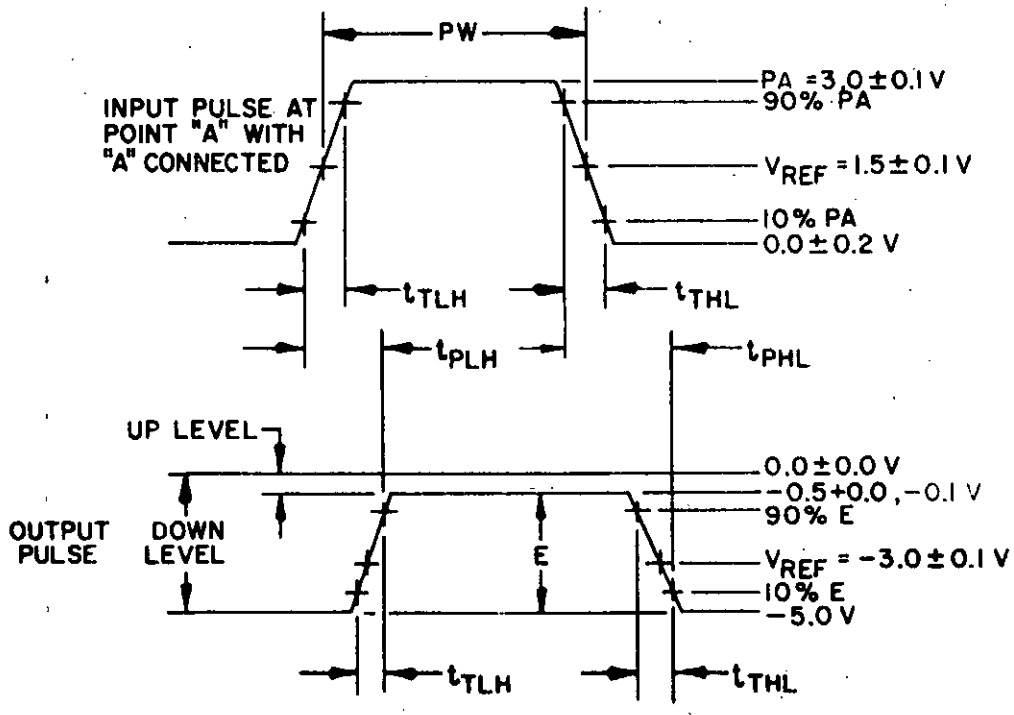
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, XV, and XVI.

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NOTES:
1. Input pulse parameters:

   PW = 1.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 500 kHz ±10%.
2. The values measured on item D05 shall comply with electrical requirements L1-L18, M1-M18, N1-N18 and P1-P18 and shall be measured per waveforms shown above. Before making these measurements, verify the output levels to be not more negative than -0.5 V for the "up" level and to be not less negative than -3.0 V for the "down" level.

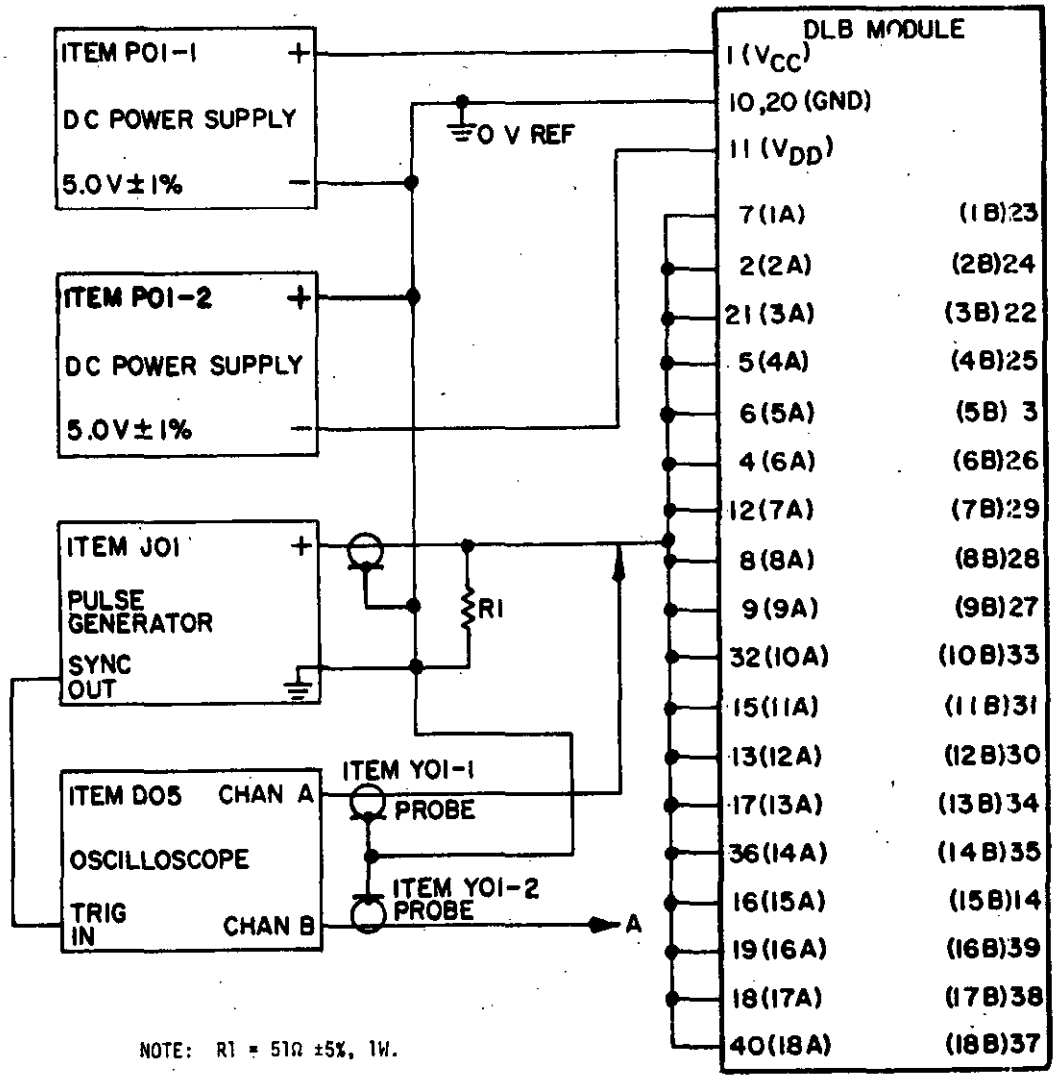
FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII, XIV, XV, and XVI.

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Power supply transient amplitude	1	0	400	mV	16, 17
Q2	Power supply transient amplitude	11	0	500	mV	16, 17

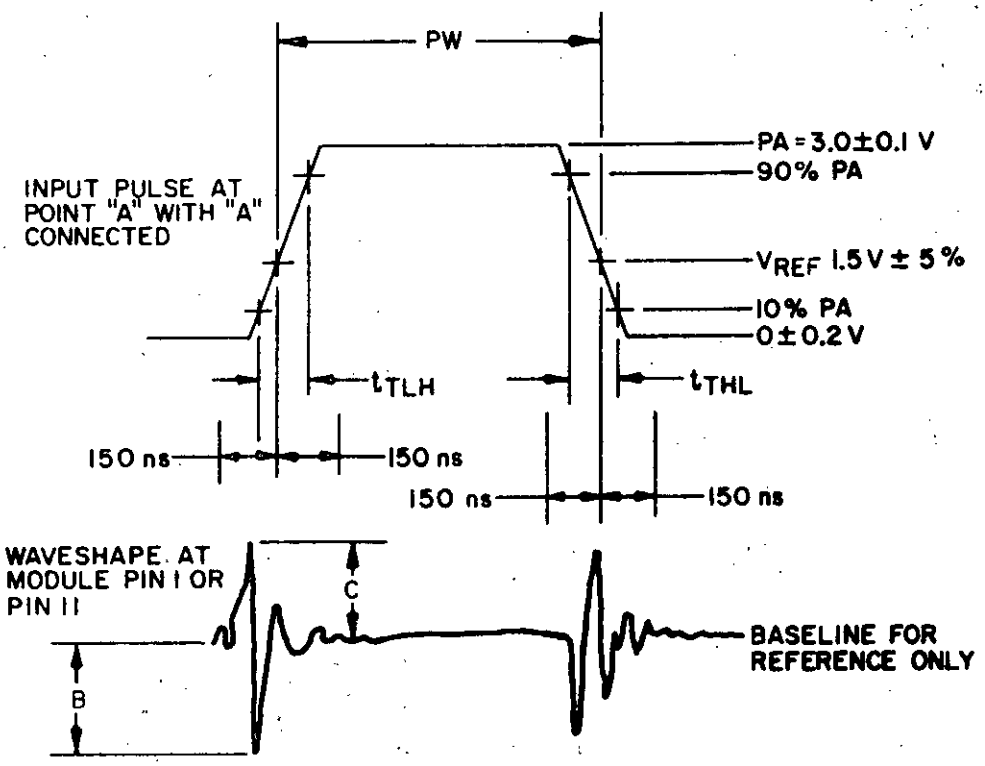
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NOTE: R1 = 51Ω ±5%, 1W.

FIGURE 16. Power supply transient amplitude test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 1.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 500 kHz ±10%.
2. Observe the power supply transient at module pin 1 on item D05. The typical waveform is shown above.
3. Amplitude "B" shall comply with electrical requirement Q1.
4. Amplitude "B" is measured from the baseline to the most negative excursion of the waveform, which may occur in time coincidence ±150 ns with the 1.5 Vdc point of either the leading edge or the trailing edge of the input waveform.
5. Observe the power supply transient at module pin 11 on item D05. The typical waveform is shown above.
6. Amplitude "C" shall comply with electrical requirement Q2.
7. Amplitude "C" is measured from the baseline to the most positive excursion of the waveform, which may occur in time coincidence ±150 ns with the 1.5 Vdc point of either the leading edge or the trailing edge of the input waveform.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level) test.

Test condition		Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1B	Output crosstalk, low level	2,21,5,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	Open	7	23	0	400	mV	18, 19
R2	2B		7,21,5,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40		2	24				
R3	3B		7,2,5,6,4, 12,8,9,32, 15,13,17.36, 16,19,18,40		21	22				
R4	4B		7,2,21,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40		5	25				
R5	5B		7,2,21,5,4, 12,8,9,32, 15,13,17,36, 16,19,18,40		6	3				
R6	6B		7,2,21,5,6, 12,8,9,32, 15,13,17,36, 16,19,18,40		4	26				
R7	7B		7,2,21,5,6, 4,8,9,32,15, 13,17,36,16, 19,18,40		12	29				

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R8	8B	Output crosstalk, low level	7,2,21,5,6, 4,12,9,32, 15,13,17,36, 16,19,18,40	Open	8	28	0	400	mV	18, 19
R9	9B		7,2,21,5,6, 4,12,8,32, 15,13,17,36, 16,19,18,40		9	27				
R10	10B		7,2,21,5,6, 4,12,8,9, 15,13,17,36, 16,19,18,40		32	33				
R11	11B		7,2,21,5,6, 4,12,8,9,32, 13,17,36,16, 19,18,40		15	31				
R12	12B		7,2,21,5,6, 4,12,8,9,32, 15,17,36,16, 19,18,40		13	30				
R13	13B		7,2,21,5,6, 4,12,8,9,32, 15,13,36,16, 19,18,40		17	34				
R14	14B		7,2,21,5,6, 4,12,8,9,32, 15,13,17,16, 19,18,40		36	35				

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
R15	15B Output crosstalk, low level	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 19,18,40	Open	16	14	0	400	mV	18, 19
R16	16B	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 16,18,40		19	39				
R17	17B	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 16,19,40		18	38				
R18	18B	7,2,21,5,6 4,12,8,9,32, 15,13,17,36, 16,19,18		40	37				

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TABLE VIII. Output crosstalk, high level, QPSK.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test figure
S1	1B Output crosstalk, high level	2,21,5,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	7	Open	23	0	400	mV	18, 19
S2	2B	7,21,5,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	2		24				
S3	3B	7,2,5,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	21		22				
S4	4B	7,2,21,6,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	5		25				
S5	5B	7,2,21,5,4, 12,8,9,32, 15,13,17,36, 16,19,18,40	6		3				
S6	6B	7,2,21,5,6, 12,8,9,32, 15,13,17,36, 16,19,18,40	4		26				
S7	7B	7,2,21,5,6, 4,8,9,32,15, 13,17,36,16, 19,18,40	12		29				

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TABLE XIX. Output crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
S8	8B	Output crosstalk, high level	7,2,21,5,6, 4,12,9,32, 15,13,17,36, 16,19,18,40	8	Open	28	0	400	mV	18, 19
S9	9B		7,2,21,5,6, 4,12,8,32, 15,13,17,36, 16,19,18,40	9		27				
S10	10B		7,2,21,5,6, 4,12,8,9, 15,13,17,36, 16,19,18,40	32		33				
S11	11B		7,2,21,5,6, 4,12,8,9,32, 13,17,36,16, 19,18,40	15		31				
S12	12B		7,2,21,5,6, 4,12,8,9,32, 15,17,36,16, 19,18,40	13		30				
S13	13B		7,2,21,5,6, 4,12,8,9,32, 15,13,36,16, 19,18,40	17		34				
S14	14B		7,2,21,5,6, 4,12,8,9,32, 15,13,17,16, 19,18,40	36		35				

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TABLE XIX. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S15	15B Output crosstalk, high level	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 19,18,40	16	Open	14	0	400	mV	18, 19
S16	16B	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 16,18,40	19		39				
S17	17B	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 16,19,40	18		38				
S18	18B	7,2,21,5,6, 4,12,8,9,32, 15,13,17,36, 16,19,18	40		37				

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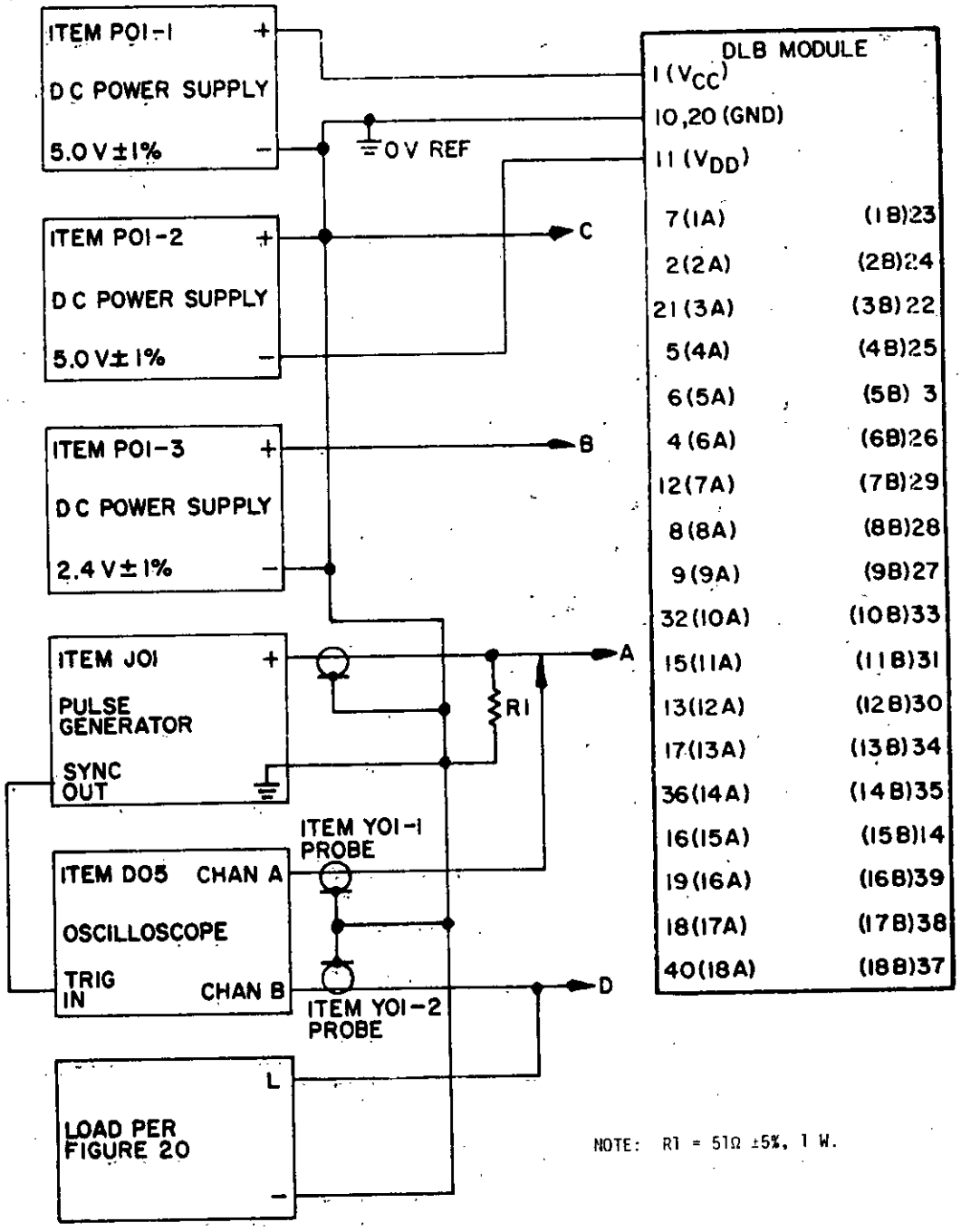
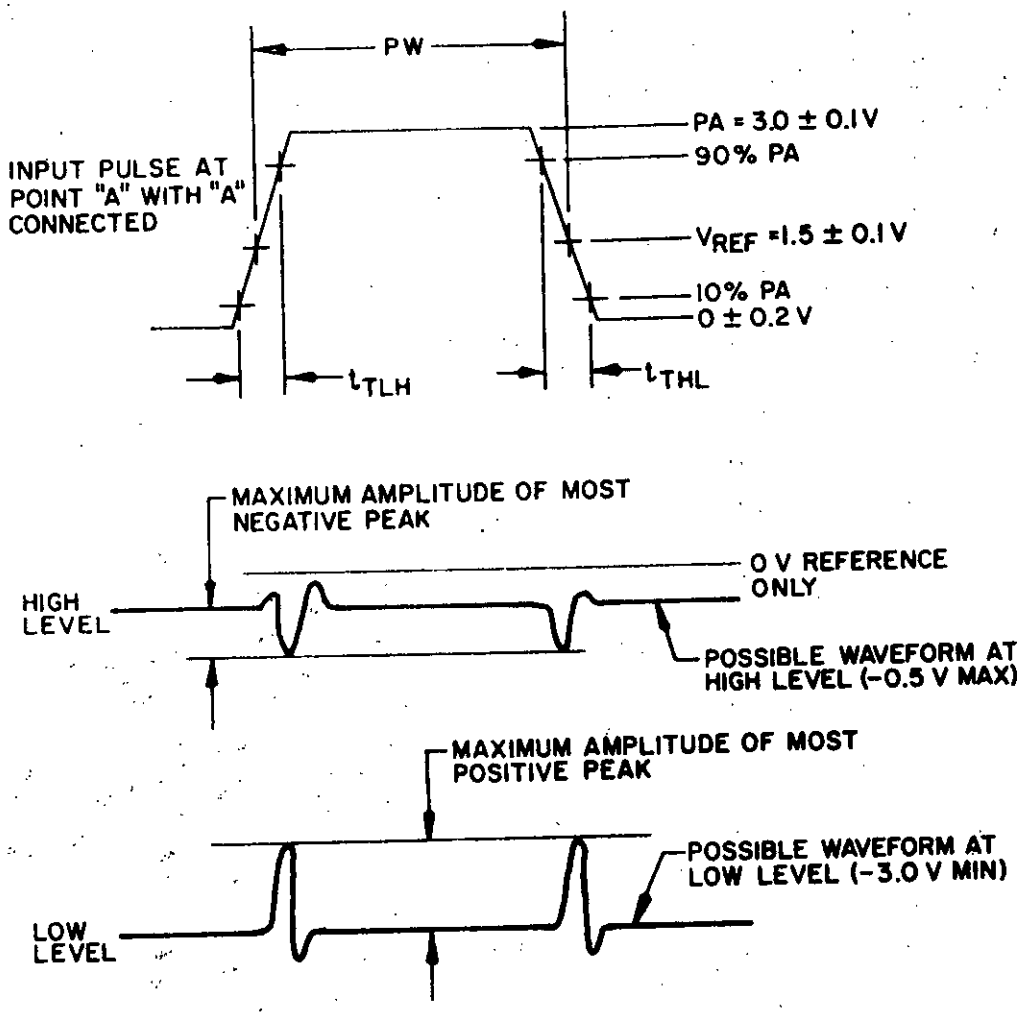


FIGURE 18. Output crosstalk (low-level and high-level) test circuit for tables XVIII and XIX.

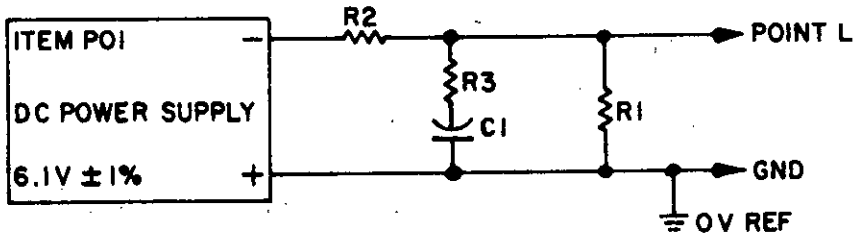
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NOTES:
1. Input pulse parameters:
   PW = 1.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 25 ns ±10%.
   PRR = 500 kHz ±10%.
2. Only the output under test shall be loaded.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVIII and XIX.

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NOTES:
1. C1 = 0.01 µF ±10% (including probe and parasitic capacitance).
2. R1 = 3.01 kΩ ±1%, 1/4 W.
3. R2 = 5.11 kΩ ±1%, 1/4 W.
4. R3 = 100Ω ±1%, 1/4 W.
5. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
6. During output switching, item POI shall have a maximum overshoot of 500 mV.

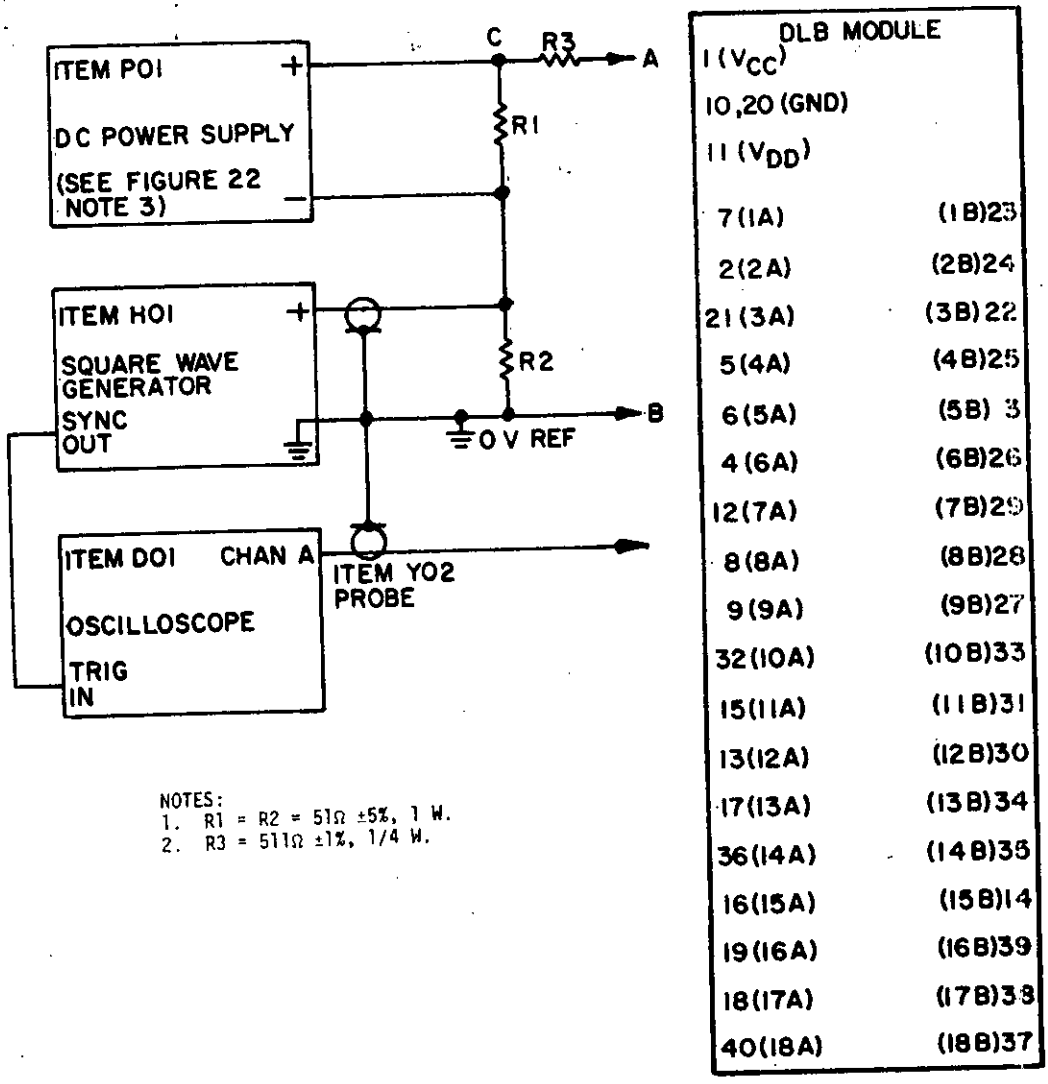
FIGURE 20. DLB logic load circuit for output crosstalk test.

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TABLE XX. Power supply transient rise time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply transient rise time	1	10, 20	2.0	5.0	ms	21, 22

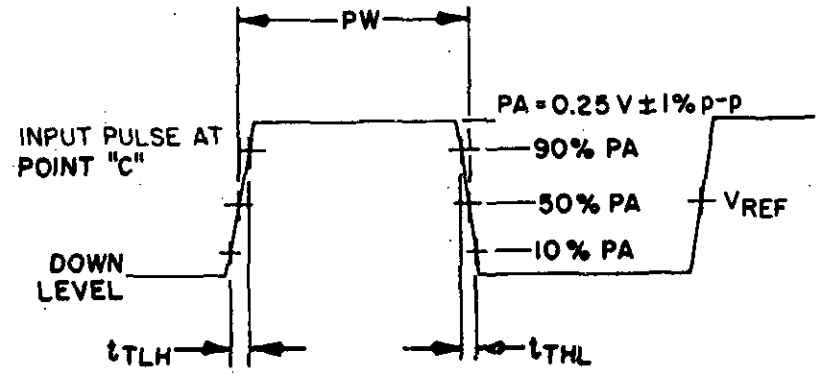
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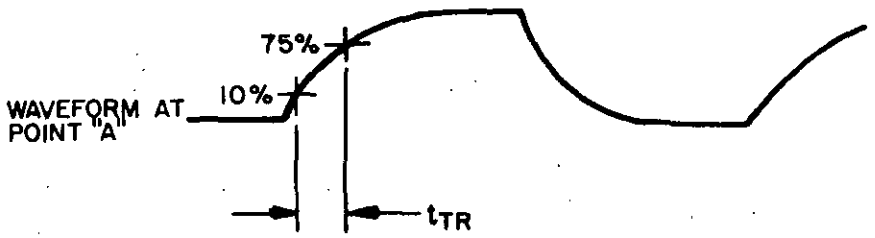


NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 21. Power supply transient rise time test circuit for table XX.

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NOTES:
1. Input pulse parameters:
   PW = 50 ms ±10%.
   $t_{TLH} = t_{THL} = \leq 100$ ns (reference).
   PRR = 10 Hz (reference).
2. With point "A" open, adjust input as shown at point "C".
3. With point "A" open, adjust item P01 such that the down level of the input is at 0 ±0.01 V.
4. Connect points "A" and "B" to module pins defined in table XX, recheck down level and pulse parameters at point "C".
5. Measure transient rise time (t_{TR}) at point "A".

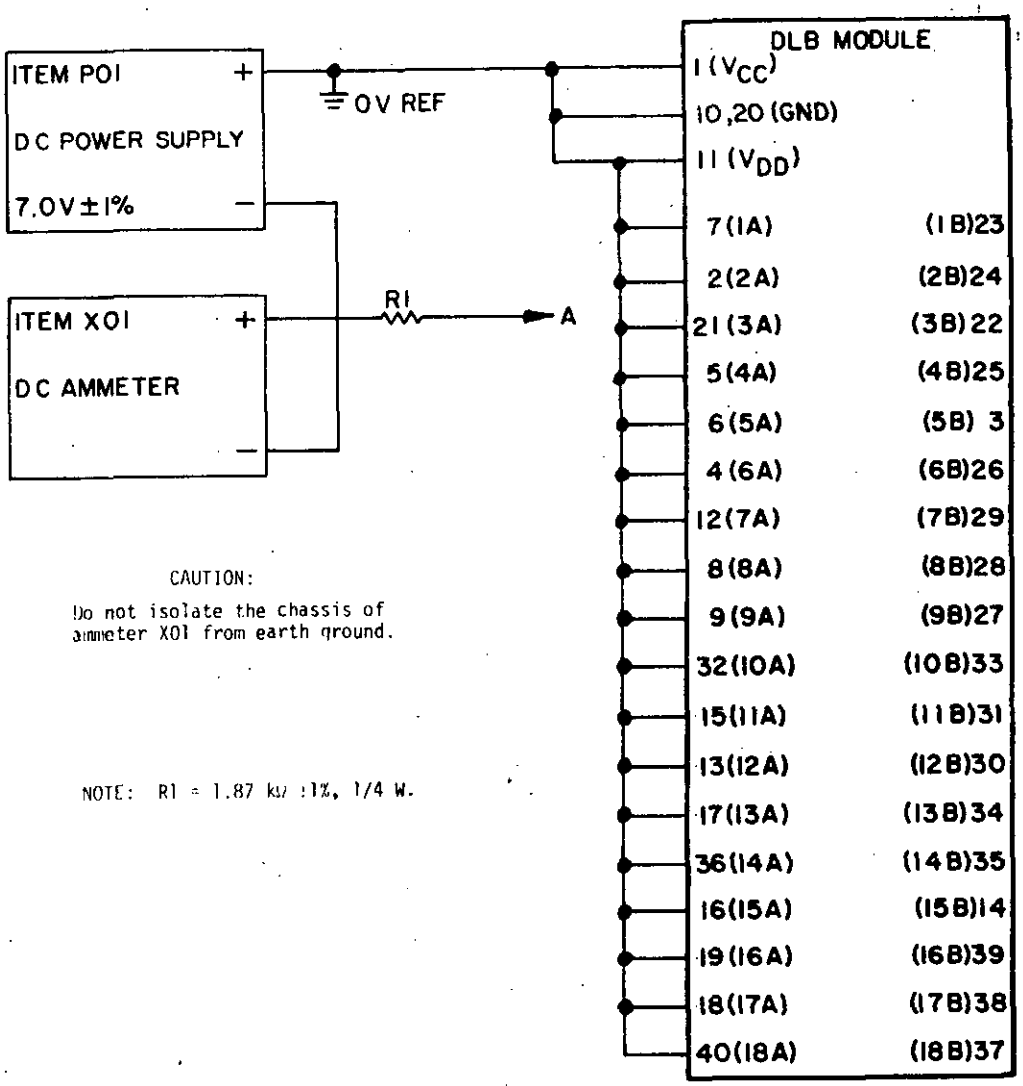
FIGURE 22. Power supply transient rise time test circuit waveforms for table XX.

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TABLE XXI. Output impedance check test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
U1	1B Output impedance check	23	0	70	µA	23
U2	2B	24				
U3	3B	22				
U4	4B	25				
U5	5B	3				
U6	6B	26				
U7	7B	29				
U8	8B	28				
U9	9B	27				
U10	10B	33				
U11	11B	31				
U12	12B	30				
U13	13B	34				
U14	14B	35				
U15	15B	14				
U16	16B	39				
U17	17B	38				
U18	18B	37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 1.87 kΩ ±1%, 1/4 W.

FIGURE 23. Output impedance check test circuit for table XXI.

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